May 2026

CSE 460: SOFTWARE ANALYSIS AND DESIGN

Homework #2 — Heart Health Imaging and Recording System

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Item 1 Working Code

The complete implementation of the Heart Health Imaging and Recording System is written in Java with a JavaFX GUI. The source code is hosted on GitHub in *HeartHealthImagingSystem* folder. The repository contains the full project including all source files, the build configuration, and sample data files demonstrating each of the five functional areas:

- **Login UI** — role-based authentication for staff and patient ID login
- **Patient Intake** — register new patients and schedule CT scan appointments
- **CT Scan Technician** — enter and save Agatston CAC score results
- **Patient View** — patients log in with their 5-digit ID to view CT results
- **Doctor View** — heart specialist reviews results, determines risk, and emails patient

GitHub repository link (code):

https://github.com/jumperpc3000/cse460-software-analysis-design/tree/main/Assignments/HW2

Item 2 Demo Video

A video demonstration is provided showing the full system in action. The demo covers:

a. How to run the project and login

b. Patient Intake functionality — entering patient data and scheduling a scan

c. CT Scan Tech functionality — recording Agatston CAC scores

d. Patient View functionality — logging in by patient ID and viewing results

e. Doctor View functionality — reviewing results, risk assessment, and sending the patient email

Demo video link (Zoom, Passcode: `2x=RLS^k`**):**

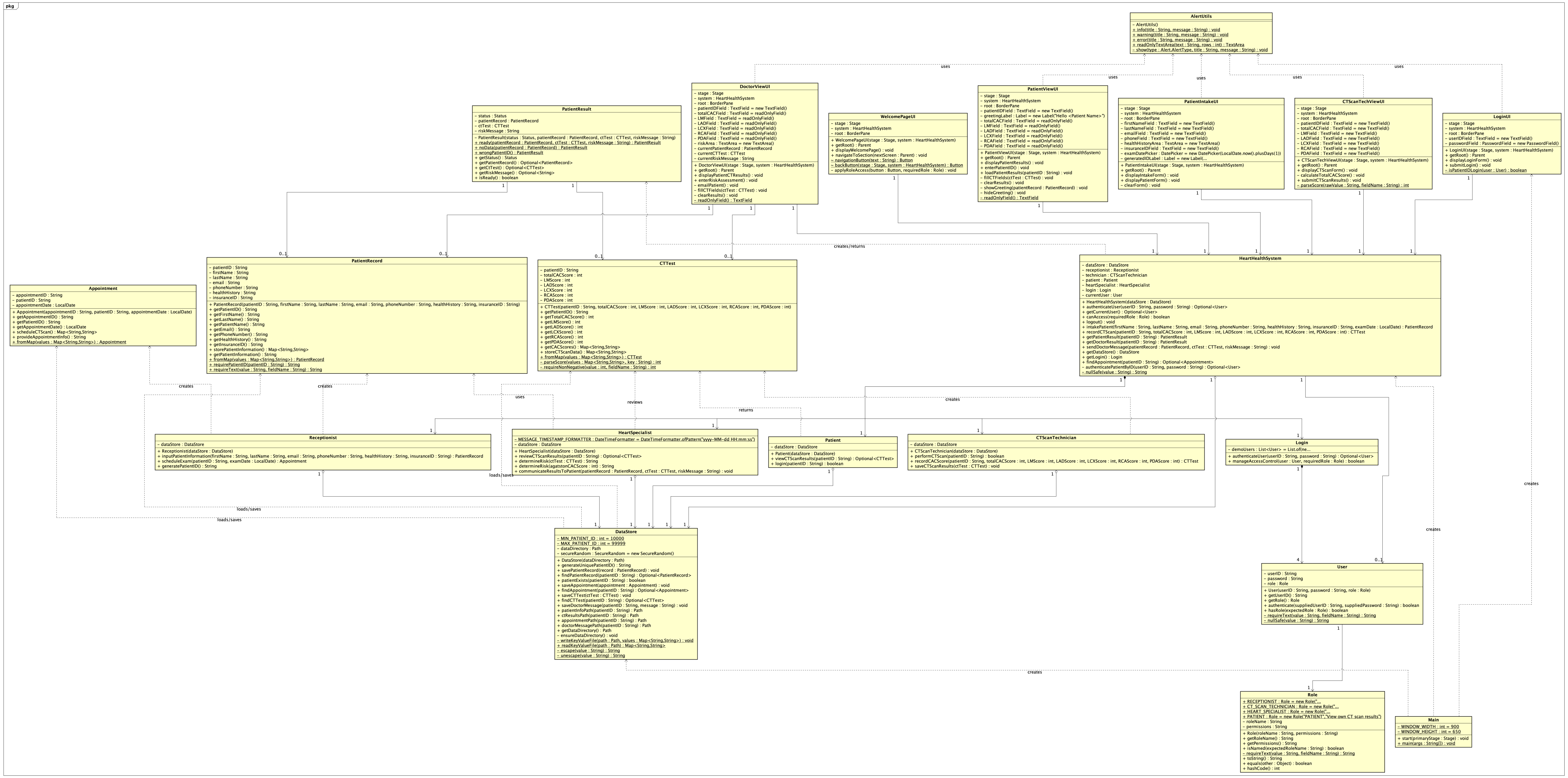
https://asu.zoom.us/rec/share/1QsCxLYSdcb4WBvU6xDsbhiIWnK3uIGY6tdskKxIWbIvQpU5B5xBJQaw7auGo42Y.ggwMa-QIWx-9ml8oPasscode:2x=RLS^k

Item 3 Final Implementation Class Diagram

The final class diagram reflecting the completed implementation is presented on the following page. It was produced in **Astah** and captures all packages, classes, fields, methods, and relationships of the delivered system.

CSE 460 Website (Assignments):

https://jumperpc3000.github.io/cse460-software-analysis-design/#assignments


pkg
AlertUtils
DoctorViewUI
PatientResult
WelcomePageUI
PatientViewUI
PatientIntakeUI
CTScanTechViewUI
LoginUI
PatientRecord
CTTest
HeartHealthSystem
Appointment
Receptionist
HeartSpecialist
Patient
CTScanTechnician
Login
DataStore
User
Role
Main
uses
creates/returns
creates
reviews
returns
loads/saves